From Consumption to Creation

A Two-Layer Framework for China's Domestic Economic Transition in an Age of Automation and Cheap Energy

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Author disclosure

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Executive Summary

China has already recognized many of the immediate requirements for strengthening domestic demand: higher household incomes, stronger social protection, expansion of services, development of new forms of consumption, technological upgrading, and better coordination between new demand and new supply.

The deeper question may be different.

As energy, robotics and artificial intelligence become progressively cheaper and more capable, an increasing share of physical production may require less human labor. At the same time, the relative importance of proximity to final markets, logistics, inventory and lead time may increase, potentially changing the geography of future investment and production.

Under such conditions, an economy cannot rely indefinitely on the traditional chain:

investment → production capacity → employment → household income → consumption → further production.

The challenge becomes how to convert rising productivity into a continuously expanding range of economically valuable human participation.

This paper proposes a two-layer transition framework:

1. Stabilisation: repair household and institutional balance sheets and reduce excessive precautionary behavior.
2. Expansion of valuable human participation: create mechanisms through which technological productivity is converted into new economic activity, markets, income and further productive capability.

The second layer can be understood as a three-level mechanism:

circulation → exchange → creation

- Circulation: ensure that part of existing monetary liquidity continues to circulate rather than being held indefinitely.
- Exchange: reduce situations in which real economic resources cannot be exchanged merely because conventional monetary liquidity is temporarily mismatched.
- Creation: continuously develop genuinely valuable new activities and markets capable of absorbing productivity gains.

Government can play an important role at the beginning of this process as an anchor customer and market catalyst, especially where suppliers and customers cannot coordinate because the market does not yet exist. But public demand should come with a predefined exit mechanism: declining public financing, measurable private-demand milestones, and explicit continuation, modification or termination criteria.

The objective is not permanent government stimulation, artificial consumption, or replacement of conventional money.

The objective is to develop an economic system capable of converting increasing productivity into new forms of valuable human participation.

1. The Problem

China's previous growth model relied heavily on a reinforcing cycle:

investment → capacity → employment → income → consumption → investment.

Exports and integration into global manufacturing chains provided an additional mechanism for absorbing the resulting production capacity.

That model generated enormous productive capabilities. However, several parts of the cycle are now under pressure.

In the near term, the relevant constraints include:

- the prolonged adjustment of the property sector;
- pressure on local-government finances;
- balance-sheet problems in parts of the financial and corporate sectors;
- high household precautionary saving;
- weak confidence in some segments of the economy;
- industrial overcapacity and intense price competition in selected sectors;
- continued dependence on external markets for absorbing some production capacity.

These problems can be addressed through conventional macroeconomic and social-policy instruments.

But there is also a longer-term structural question.

If robotics, artificial intelligence and energy technologies continue to reduce the cost of production, fewer human hours may be required to produce the same quantity of physical goods.

This creates a potential discontinuity in the traditional economic mechanism.

An economy can become increasingly capable of producing goods while becoming less dependent on human labor for their production.

The question then changes from:

> *How can society produce more?*

to:

> *How can society continuously create new forms of economically valuable activity as production becomes more automated?*

This is not primarily a question of increasing consumption for its own sake.

It is a question of expanding the set of things that people and organizations can do that others genuinely value.

2. What China Already Knows

The proposed framework does not assume that China has overlooked the immediate domestic-demand problem.

China's current policy direction already includes many relevant elements:

- strengthening domestic demand;
- increasing household income;
- improving social protection;
- expanding services;
- developing new forms of consumption;
- encouraging new demand and new supply;
- promoting artificial intelligence and robotics;
- developing lifelong learning and new occupations;
- improving the efficiency of government investment;
- using government purchasing and investment where appropriate.

The important distinction is therefore not between:

"existing Chinese policy"

and

"this proposal."

Instead, the proposal attempts to provide an additional causal architecture connecting several existing policy directions.

In particular, it asks how:

balance-sheet stabilisation → productivity growth → distribution of productivity gains → new economic activity → private demand → further activity

can become a self-reinforcing process.

The framework therefore should be viewed as complementary to conventional macroeconomic, fiscal, industrial and social-policy analysis rather than as a substitute for it.

3. Structural Scenario: Cheap Energy, Cheap Robots and Proximity to Demand

The following is a scenario rather than a forecast.

Suppose that over the next two decades:

- energy becomes substantially cheaper and more abundant;
- industrial robots become substantially cheaper and more capable;
- artificial intelligence reduces the cost of coordination, engineering, design and knowledge work;
- automated production requires progressively less human labor;
- manufacturing becomes increasingly flexible and geographically distributed.

The economic geography of production could then change.

Historically, low labor costs were one of the strongest reasons for concentrating manufacturing far from final consumers.

As labor becomes a smaller component of total production cost, other factors can become relatively more important:

- logistics;
- inventory;
- delivery time;
- supply-chain resilience;
- proximity to customers;
- energy availability;
- local infrastructure;
- regulatory conditions.

This could encourage partial regionalisation or reshoring of some production, particularly for products where automation makes labor-cost differences less important.

This does not imply that manufacturing will simply leave China.

China's existing advantages include industrial ecosystems, infrastructure, supplier density, engineering capabilities, accumulated capital and enormous domestic demand.

The point is different:

> *The economic value of low-cost labor may decline faster than the value of the broader industrial ecosystem.*

China may therefore face a future in which maintaining economic activity requires more than maintaining production capacity.

It may require continuously generating new forms of value in which human participation remains economically meaningful.

4. The Core Hypothesis

The central hypothesis of this paper is:

> *As production becomes increasingly automated, the strategic constraint on a large economy may gradually shift from its capacity to produce goods toward its capacity to generate economically valuable human participation.*

This does not mean preserving existing jobs regardless of economic value.

It means expanding the range of activities in which human time, knowledge, creativity, care, judgment, social interaction and entrepreneurship can generate value.

Examples may include:

- scientific research;
- healthcare;
- elderly and childcare;
- education;
- environmental restoration;
- culture;
- sports;
- complex engineering;
- exploration;
- personalised services;
- new forms of digital and physical experience;
- robotics-enabled household services;
- scientific and technical entrepreneurship.

The objective is not to manufacture demand for activities people do not value.

The objective is to discover and develop activities that people and organisations voluntarily value enough to support economically.

This distinction is essential.

5. Two-Layer Architecture

Layer A — Stabilisation

The first layer addresses the existing economic imbalance.

Relevant mechanisms include:

Household balance sheets

- stabilising the property adjustment;
- strengthening pensions, healthcare and unemployment protection;
- improving long-term-care and childcare provision;
- improving access to basic public services;
- addressing remaining barriers affecting migrant and rural households;
- reducing excessive precautionary saving caused by uncertainty.

Institutional balance sheets

- addressing local-government financial constraints;
- improving the quality and efficiency of public investment;
- strengthening financial-sector resilience;
- increasing predictability for private enterprises.

Fiscal orientation

Where appropriate, a greater share of fiscal resources can support households and services rather than simply expanding physical production capacity.

The objective is not merely to increase consumption.

It is to restore the transmission mechanism between:

income → confidence → demand → productive activity.

Layer B — Expansion of Valuable Human Participation

Stabilisation alone does not answer the longer-term automation problem.

The second layer seeks to establish a new loop:

human capability and insight → unmet or newly recognised needs → new products, services or experiences → voluntary private demand → income and investment → greater capability → further valuable activity

Education, vocational adaptation and AI tools are important here, but they are amplifiers rather than the foundation.

Education creates capability.

AI can multiply capability.

Robotics can free resources.

But sustainable economic activity ultimately requires something else:

someone must value the resulting activity enough to support it.

6. A Three-Level Mechanism

The transition can be viewed as three connected problems.

Level 1 — Circulation

An economy may possess substantial financial resources while some of those resources remain inactive or are primarily used as stores of value.

One possible experimental mechanism is a designated circulation instrument with a carrying cost or demurrage mechanism.

Under such a system, the instrument loses some of its holding value over time unless it is spent or otherwise returned to circulation.

The purpose would not be to replace conventional money.

Instead:

- conventional CNY remains available for saving, investment and external transactions;
- a limited experimental instrument is used for specific domestic circulation purposes;
- its carrying cost increases the incentive to circulate it;
- the effect can be measured against a control group or comparable economic activity.

Historical local experiments with demurrage-style currencies provide evidence that carrying costs can affect short-term circulation behaviour.

They do not establish that such a mechanism is viable at national scale.

Potential problems include:

- arbitrage;
- currency substitution;
- inflationary effects;
- administrative complexity;
- incentives to convert the instrument into conventional assets;
- interaction with existing monetary policy.

Therefore this should be treated as a controlled experiment, not as a replacement for the monetary system.

A programmable digital currency infrastructure such as e-CNY could potentially provide a technical basis for testing such a designated instrument, subject to monetary, legal and institutional analysis.

7. Level 2 — Exchange

The second problem is different.

Sometimes real economic resources exist, but conventional monetary liquidity is temporarily mismatched.

Consider four participants:

- A has unused machine capacity;
- B has specialised labour;
- C has materials;
- D has customers or final demand.

In a conventional system, each transaction may require sufficient cash liquidity at the correct point in the chain.

A temporary shortage at one point can prevent several otherwise viable transactions from occurring.

A mutual-credit or multilateral-clearing system can address this differently.

Instead of requiring every participant to settle every transaction independently in conventional money, participants can accumulate reciprocal obligations that are cleared multilaterally.

For example:

A provides machine capacity to B → B provides services to C → C provides materials to D → D pays for the final product → the network clears the corresponding obligations.

The system does not create resources from nothing.

It attempts to make better use of resources that already exist but are poorly connected by conventional liquidity.

This is not a replacement for money and not simply barter.

It is an additional clearing layer.

Potential benefits:

- lower dependence on point-in-time cash liquidity;
- better utilisation of idle capacity;
- support for SME networks;
- improved matching of complementary resources.

Potential risks:

- valuation problems;
- default;
- hidden leverage;
- governance capture;
- concentration of clearing power;
- fraud;
- systemic propagation of local failures.

Therefore mutual credit should also begin as a controlled experimental layer.

8. Level 3 — Creation

The first two levels are mechanisms for improving the use of existing economic resources.

They are not the ultimate objective.

The third level is the most important:

> *What new things can society create that people genuinely value?*

If productivity rises while the quantity of economically valuable activity does not expand correspondingly, the system may experience a growing gap between:

capacity to produce

and

capacity to generate economically valuable demand and participation.

The response should not be to manufacture consumption.

Instead, society can expand its ability to discover and develop new markets.

Potential fields include:

- scientific and technological research;
- advanced healthcare;
- elderly and childcare;
- education and lifelong learning;

- environmental restoration;
- culture;
- sports;
- personalised services;
- complex engineering;
- exploration;
- new physical-digital experiences;
- robotics-enabled services;
- scientific and technological entrepreneurship.

Some of these activities will fail.

That is normal.

A successful transition therefore requires mechanisms for experimentation, measurement and termination rather than a requirement that every new activity succeed.

9. Government as an Initial Market Anchor — with a Defined Exit

One of the central practical problems in creating a new market is coordination.

Potential suppliers may say:

> "*We will build it when customers exist.*"

Potential customers may say:

> "*We will buy it when suppliers and infrastructure exist.*"

Investors may say:

> "*We will invest when both exist.*"

The market can therefore remain trapped in a state where everyone is waiting for everyone else.

Government procurement can temporarily break this coordination problem.

The proposed role is therefore not simply:

government → permanent buyer.

It is:

government → initial market anchor → private market formation → government withdrawal.

This can be called:

> *government-anchored market formation with a predefined exit.*

For each experiment, the exit mechanism should be established before significant public resources are committed.

A possible structure is:

Stage 1 — Public anchor

Government provides initial procurement, infrastructure, standards, guarantees or other clearly defined support.

Stage 2 — Market formation

Private suppliers and users begin to participate.

Stage 3 — Measurable transition

Specific milestones must be reached, such as:

- increasing private demand;
- increasing private-provider participation;
- voluntary household or corporate payments;
- declining dependence on public financing;
- independent investment;
- repeat purchases outside the original government programme.

Stage 4 — Public withdrawal

Public financing or procurement declines according to the predefined schedule.

Stage 5 — Evaluation

The experiment is assessed according to what happens after public support is reduced.

The key question is not:

> "*Did government spending create activity?*"

It is:

> "*Did public intervention create conditions under which independent economic activity continued?*"

Persistent public financing is not evidence that an activity has no social value.

However, if the stated objective is the creation of a self-sustaining market, continued discretionary public financing should not itself be counted as evidence that such a market has formed.

10. The Integrated Mechanism

The three levels can be represented as:

Circulation → Exchange → Creation

Circulation

A designated circulation instrument reduces the incentive to hold that particular instrument as a store of value and encourages turnover.

↓

Exchange

Mutual-credit and multilateral-clearing mechanisms allow real resources to be exchanged when conventional monetary liquidity is temporarily insufficient or mismatched.

↓

Creation

Freed and better-connected resources are directed toward genuinely valuable new activities and markets.

↓

Private demand

Successful activities generate voluntary demand, income and investment.

↓

Further creation

The resulting resources expand society's capacity to create additional valuable activity.

This is the intended long-term loop.

The financial mechanisms are therefore subordinate to the creation of real economic value.

Neither demurrage nor mutual credit is the objective.

They are experimental mechanisms for addressing particular frictions in circulation and exchange.

11. A Possible Technical Architecture

The financial mechanisms should not be interpreted as requiring a complete replacement of China's existing monetary system.

A possible experimental architecture could instead be layered:

CNY

↓

existing digital-payment infrastructure / e-CNY

↓

designated experimental circulation instrument

↓

optional programmable carrying-cost mechanism

↓

participating merchants, households and organisations

↓

optional mutual-credit / multilateral-clearing layer

↓

conversion and settlement in conventional CNY

This architecture would preserve a clear distinction between:

- conventional money;
- an experimental circulation instrument;
- a clearing mechanism;
- real economic activity.

The technical architecture should be designed so that an experiment can be stopped without destabilising the broader monetary system.

Any such experiment would require independent analysis of monetary policy, taxation, accounting, consumer protection, banking regulation, cybersecurity, privacy and capital controls.

12. What Should Count as Success?

The central evaluation criteria should concern independent economic activity, rather than the volume of government-supported activity.

Possible indicators include:

Market formation

- number of independent private providers;
- number of independent buyers;
- repeat voluntary purchases;
- private investment;
- price discovery.

Public-support dependence

- share of revenue originating from government procurement;
- public financing per unit of economic activity;
- change in that share over time.

Productivity absorption

Measurement of whether previously underutilised:

- human time;
- knowledge;
- capital;
- infrastructure;
- automated production capacity

is converted into additional measurable economic value.

Persistence

The critical test:

> *What happens after public support is reduced?*

An experiment that continues or expands under declining public support has demonstrated a different economic property from one that requires permanent public financing.

13. Risks and Correction Triggers

The framework contains significant risks.

Bureaucratic capture

Government programmes can become permanent institutions whose survival becomes the objective.

Correction: predefined termination or redesign criteria.

Artificial markets

Public procurement can create activity that disappears immediately when government stops paying.

Correction: measure private demand independently of public procurement.

Credential inflation

Expansion of education can produce credentials without corresponding productive value.

Correction: evaluate outcomes in terms of capabilities and economic activity rather than qualifications alone.

Financial instability

Experimental mutual-credit systems can create hidden leverage or interconnected defaults.

Correction: strict exposure limits, transparent accounting and controlled pilots.

Regional concentration

New economic activity may concentrate in already-developed regions.

Correction: monitor geographic distribution and access to infrastructure.

Unequal distribution of productivity gains

Automation can increase output while concentrating income and ownership.

Correction: evaluate how productivity gains are distributed across households, enterprises, investment and public resources.

Automation outrunning adaptation

Technology may eliminate economically valuable roles faster than new roles and markets emerge.

Correction: monitor the pace of labour displacement and the rate of creation of new economically valuable activity.

Permanent state dependence

A programme may become dependent on continued government purchasing.

Correction: declining public-support schedule and explicit private-demand milestones established before launch.

14. Experimental Governance

The proposed framework should operate as a portfolio of experiments rather than as a single nationwide reform.

Each experiment should specify in advance:

1. Problem being addressed
2. Target population
3. Initial public intervention
4. Expected private-demand mechanism
5. Economic resources to be activated
6. Measurable indicators
7. Maximum public commitment
8. Private-demand milestones
9. Exit schedule
10. Termination or redesign criteria

A practical evaluation period could initially be approximately 12–24 months, depending on the type of experiment.

The important principle is:

> *No experiment should be judged solely by activity created while public support is rising.*

The stronger test is what happens when public support stops increasing and begins to decline.

15. What This Framework Does Not Claim

This paper does not claim that:

- China should immediately replace conventional monetary policy;
- demurrage money has been proven at national scale;
- mutual credit can replace banking;
- government can manufacture sustainable demand;

- automation will necessarily cause mass unemployment;
- manufacturing will necessarily leave China;
- cheap energy is guaranteed;
- reshoring will occur uniformly across industries;
- every new service represents genuine economic value;
- government procurement is sufficient to create a market.

The framework is instead a systems-engineering hypothesis.

Its purpose is to identify mechanisms that can be tested against economic reality.

16. Summary

China does not necessarily have to choose between production and consumption.

The deeper challenge may be that production itself is becoming sufficiently efficient that the traditional relationship between:

productivity → employment → income → demand

can weaken.

The transition can therefore be viewed as a three-level problem.

First, existing liquidity must circulate sufficiently.

Second, viable economic exchange should not be prevented solely by temporary shortages or mismatches of conventional monetary liquidity.

Third, society must continuously create new activities that generate genuine economic value.

This produces the following architecture:

Stabilisation

→ healthier household and institutional balance sheets

Circulation

→ better utilisation of existing monetary liquidity

Exchange

→ better utilisation of complementary real resources

Creation

→ new valuable products, services, research, experiences and activities

Private demand

→ independent economic sustainability

Further productivity and capability

→ further creation.

Government can temporarily act as an anchor customer and market catalyst to overcome the initial coordination problem.

But every such intervention should contain a predefined path toward:

declining public support → increasing private demand → independent economic activity.

The objective is not permanent state stimulation, faster consumption for its own sake, or replacement of conventional money.

It is the creation of an economic system capable of converting rising productivity into new forms of valuable human participation.

The fundamental question is therefore not:

> *How can an economy preserve a system in which people are needed to produce everything society consumes?*

It is:

> *How can an economy build institutions and markets in which increasing automation frees human capacity faster than it destroys the economic structures through which that capacity can create value?*

About the Author and Why This Can Be Evaluated on Its Merits

Dmitry Kazakov is an independent Software & Systems Engineer with professional software-development experience since 2010 and independent software and systems-engineering practice since 2015.

His work spans software, data systems, hardware-adjacent applications, security and industrial systems, as well as technology-oriented businesses and products. His approach combines systems engineering with economic and organisational analysis, with particular attention to interacting constraints, feedback mechanisms and testable system architectures.

This paper is an independent analytical and systems-engineering construct. It does not represent the policy of any government, company, research institution or other organisation.

It is not presented as a formal academic macroeconomic model. Its purpose is to provide a mechanism that can be examined alongside econometric, monetary, fiscal and policy analysis.

The framework can be evaluated on its merits through questions such as:

- Is the causal mechanism internally coherent?
- Are the proposed experiments measurable?
- Can public intervention be distinguished from genuine private demand?
- Are the risks and failure modes explicit?
- Can the mechanisms be tested without destabilising the broader economic system?
- Does the framework provide useful additional language for discussing the transition from production-led growth toward an increasingly automated economy?

The author does not claim institutional authority over Chinese economic policy.

The purpose is more limited:

> *to offer a mechanism that can be examined, challenged, improved and, where appropriate, tested by people and institutions with the relevant economic expertise and decision-making authority.*

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Final Note

The proposal deliberately separates three different questions:

What is already happening? Existing Chinese policy and economic conditions.

What might happen? The structural scenario involving cheaper energy, robotics, AI and changing production geography.

What could be tested? The proposed mechanisms for stabilisation, circulation, exchange and creation.

Keeping these distinctions explicit is essential.

The value of the framework therefore does not depend on whether every assumption proves correct.

Its value depends on whether the proposed causal mechanisms are sufficiently clear that their assumptions, measurements and failure modes can be tested against reality.